

5.1 Chemical Energetics (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	5. Physical Chemistry (A Level Only)
Topic	5.1 Chemical Energetics (A Level Only)
Difficulty	Hard

Time allowed: 50

Score: /37

Percentage: /100

Question 1a

Potassium sulfide is a reagent used in analytical chemistry and pharmaceutical preparations.

Draw a fully labelled Born-Haber cycle for the formation of solid potassium from its elements.

Include state symbols for all species involved.

[7 marks]

Question 1b

Use the data in Table 1.1 to calculate the lattice enthalpy of potassium sulfide, ΔH^θ_{latt} . Show your working.

Table 1.1

Enthalpy change	Enthalpy change / kJ mol^{-1}
Formation of potassium sulfide	- 381
1st electron affinity of sulfur	- 200
2nd electron affinity of sulfur	+ 640
Atomisation of sulfur	+ 279
1st ionisation energy of potassium	+ 419
Atomisation of potassium	+ 89

 $\Delta H^\theta_{latt} = \dots\dots\dots \text{kJ mol}^{-1}$ **[3 marks]****Question 1c**

Explain the difference in values between the first and second electron affinities of sulfur.

[2 marks]

Question 2a

Magnesium oxide is often used in optical applications due to its light-reflecting properties in crystal form.

Write an equation to represent the lattice energy of MgO.

[1 mark]

Question 2b

Use the data in Table 2.1, to calculate a value for the second ionisation energy of magnesium. ΔH_{ie}^θ . Show your working.

Table 2.1

Enthalpy change	Enthalpy change / kJ mol^{-1}
Lattice energy of MgO (s)	- 3791
Enthalpy change of atomisation of Mg	148
Enthalpy of atomisation of oxygen	248
Electron affinity of the oxygen atom	-141
Electron affinity of the oxygen anion, O^-	798
First ionisation energy of Mg	736
Enthalpy of formation of MgO	552

second ionisation energy of Mg = kJ mol^{-1}

[3 marks]

Question 2c

Magnesium oxide is generally insoluble in water whereas calcium oxide is sparingly soluble. Explain why there is no enthalpy of solution data for calcium oxide.

[1 mark]

Question 3a

The incomplete Born-Haber cycle for sodium selenide is shown below in Fig. 3.1.

Write the equations for processes 1, 2 and 3.

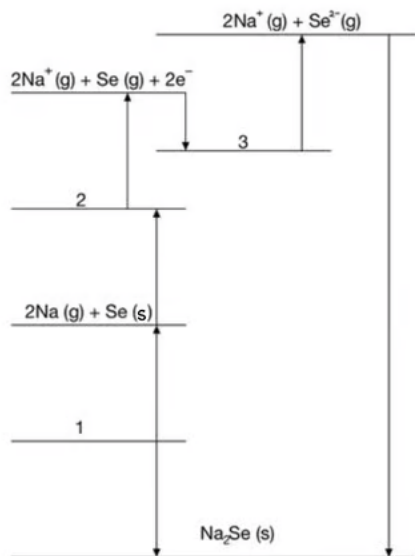


Fig. 3.1

1

2

3

[3 marks]

Question 3b

If sulfur is used as opposed to selenium in the lattice, suggest the change in value of the lattice energy, ΔH°_{latt} . Explain your answer.

Change

Explanation

.....

[3 marks]

Question 3c

Use the data in Table 3.1 to calculate the lattice energy of aluminium oxide, ΔH°_{latt} . Show your working.

Table 3.1

Enthalpy Change	Enthalpy Change / kJ mol^{-1}
Atomisation of aluminium	+326
Atomisation of oxygen	+249
First ionisation energy of aluminium	+578
Second ionisation energy of aluminium	+1817
Third ionisation energy of aluminium	+2745
Electron affinity of O atom	-141
Electron affinity of O^-	+753
Formation of aluminium oxide	-1670

$\Delta H^\circ_{latt} = \dots\dots\dots \text{kJ mol}^{-1}$

[3 marks]

Question 4a

A Group 1 metal, M, formed an ionic compound with chlorine.

The enthalpy of lattice dissociation for this compound is $+773 \text{ kJ mol}^{-1}$ and the hydration enthalpy of a chloride ion is -363 kJ mol^{-1} .

The enthalpy of solution of the Group 1 chloride is $+4 \text{ kJ mol}^{-1}$.

Using the information in Table 4.1, identify the Group 1 ion, M^+ . Show your working.

Table 4.1

Group 1 ion	Enthalpy of hydration (kJ mol^{-1})
Li^+	-519
Na^+	-406
K^+	-321
Rb^+	-296

Identity of M^+ =

[2 marks]

Question 4b

The same Group 1 metal from part (a) forms an ionic lattice with another halide ion. This new ionic compound has an enthalpy of lattice formation of -705 kJ mol^{-1} .

Using **M** to represent the Group 1 metal, suggest a formula for the new ionic lattice and explain your answer.

[4 marks]**Question 4c**

The enthalpy of hydration becomes less exothermic as you go down Group 1. Using the values in part (a):

i)

Explain why the enthalpy of hydration of Group 1 ions is negative.

[3]

ii)

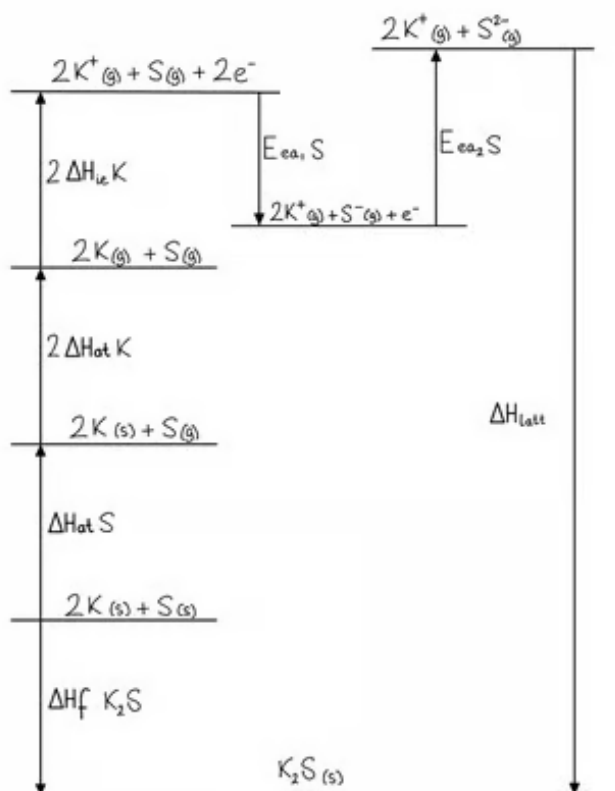
Explain why the enthalpies of hydration become less negative as you go down the group.

[2]**[5 marks]**

Model Answers: Hard

1a

a) The complete Born-Haber cycle is:



- $\Delta H_f K_2S$ AND $2K(s) + S(s)$; [1 mark]
- $\Delta H_{at} S$ AND $2K(s) + S(g)$; [1 mark]
- $2\Delta H_{at} K$ AND $2K(g) + S(g)$; [1 mark]
- $2\Delta H_{ie} K$ AND $2K^+(g) + S(g) + 2e^-$; [1 mark]
- $E_{ea1} S$ AND $2K^+(g) + S^-(g) + e^-$; [1 mark]
- $E_{ea2} S$ AND $2K^+(g) + S^{2-}(g)$; [1 mark]
- $K_2S(s)$ AND ΔH_{latt} ; [1 mark]

[Total: 7 marks]

- You must be able to draw Born Haber cycles from scratch
- It is common for harder questions to include elements that form ions with a 2+ or

2- charge so you need to make sure this is reflected in your cycle with regard to the ionisation energy and electron affinity

- **Remember:** Arrows pointing upwards represent an endothermic reaction, and down is exothermic
- Different exam boards write the symbols for enthalpy terms in different ways so don't be confused if they look slightly different in your exam, they are still correct

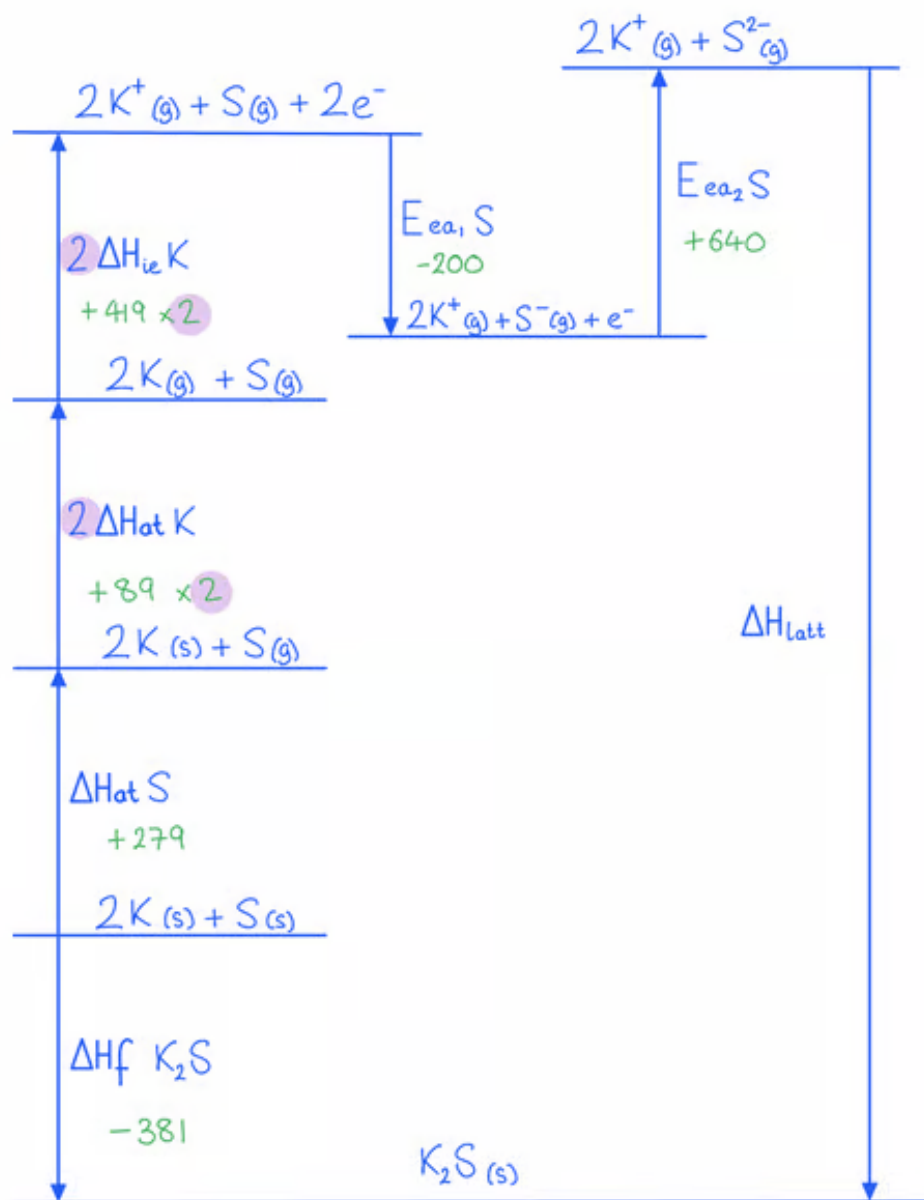
1b

b) To calculate lattice energy:

- $\Delta H_f^\theta = \Delta H_{at}^\theta(S) + 2\Delta H_{at}^\theta(K) + 2\Delta H_{ie}^\theta(K) + E_{ea1}(S) + E_{ea2}(S) + \Delta H_{latt}^\theta$; [1 mark]
- $\Delta H_{latt}^\theta = -381 - (279 + 178 + 838 - 200 + 640)$; [1 mark]
- $-2116 \text{ kJ mol}^{-1}$; [1 mark]

[Total: 3 marks]

- $\Delta H_f = \Delta H_{at}(S) + 2\Delta H_{at}(K) + 2\Delta H_{ie}(K) + E_{ea1}(S) + E_{ea2}(S) + \Delta H_{latt}$ can be simplified to
 - $\Delta H_f^\theta = \Delta H_f^\theta + \Delta H_{latt}^\theta$
 - Here ΔH_f^θ includes all the terms that convert elements from their standard state into gaseous ions so ΔH_{at}^θ , ΔH_{at}^θ , ΔH_{ie}^θ , E_{ea1} and E_{ea2}
- This can be rearranged to $\Delta H_{latt}^\theta = \Delta H_f^\theta - \Delta H_f^\theta$
- Then the numbers can be substituted in
- Don't forget to multiply the enthalpy of atomisation and ionisation energy of potassium by 2 as there are two moles present
- You can also use your Born-Haber cycle by starting at the arrow for the enthalpy change you want to find and going around the cycle in the other direction
 - Don't forget to change the sign if it points in the opposite direction from the one you are going in
 - $\Delta H_{latt}^\theta = -(+640) - (-200) - (2 \times 419) - (89 \times 2) - (+279) + (-381) = -2116 \text{ kJ mol}^{-1}$



1c

c) There is a difference because:

- (The value for) the first electron affinity is exothermic
AND
The second electron affinity is endothermic; [1 mark]
- Because there is repulsion between the negative ion and negative electron; [1 mark]

[Total: 2 marks]

- The first electron affinity is the enthalpy change when one mole of gaseous atoms forms one mole of negative ions with a single negative charge
 - Energy is released so it is exothermic
- The second electron affinity is the enthalpy change when one mole of gaseous ions with a single negative charge forms one mole of gaseous ions with a double negative charge
 - Energy is taken in to overcome the repulsion between the negative ion and negative electron
- It is common to ask about the second electron affinity for oxygen or sulfur so make sure you learn the definitions

2a

a) An equation to represent the lattice energy of MgO is:

- $\text{Mg}^{2+}(\text{g}) + \text{O}^{2-}(\text{g}) \rightarrow \text{MgO}(\text{s})$; [1 mark]

[Total: 1 mark]

- Lattice energy is the enthalpy change when 1 mole of magnesium chloride is formed from gaseous magnesium and chloride ions
- The equation for this process must include correct state symbols and charges

2b

b) To calculate the second ionisation energy of magnesium. $\Delta H^\theta_{\text{Mg}_{ie2}}$:

- $\Delta H^\theta_f = \Delta H^\theta_{at}(\text{Mg}) + \Delta H^\theta_{ie1}(\text{Mg}) + \Delta H^\theta_{ie2}(\text{Mg}) + \Delta H^\theta_{at}(\text{O}) + E_{ea1}(\text{O}) + E_{ea2}(\text{O}) + \Delta H^\theta_{latt}(\text{MgO})$; [1 mark]
- $\Delta H^\theta_{ie2}(\text{Mg}) = -552 - 148 - 736 - 248 + 141 - 798 + 3791$; [1 mark]
- $\Delta H^\theta_{ie2}(\text{Mg}) = 1450 \text{ (kJ mol}^{-1}\text{)}$; [1 mark]

[Total: 3 marks]

- Take these calculations step by step
 - $\Delta H^\theta_f = \Delta H^\theta_{at}(\text{Mg}) + \Delta H^\theta_{ie1}(\text{Mg}) + \Delta H^\theta_{ie2}(\text{Mg}) + \Delta H^\theta_{at}(\text{O}) + E_{ea1}(\text{O}) + E_{ea2}(\text{O}) + \Delta H^\theta_{latt}(\text{MgO})$
- Becomes:
 - $\Delta H^\theta_{ie2}(\text{Mg}) = \Delta H^\theta_f - [\Delta H^\theta_{at}(\text{Mg}) + \Delta H^\theta_{ie1}(\text{Mg}) + \Delta H^\theta_{at}(\text{O}) + E_{ea1}(\text{O}) + E_{ea2}(\text{O}) + \Delta H^\theta_{latt}(\text{MgO})]$

OR

$$\Delta H_{\text{ie2}}^{\ominus}(\text{Mg}) = -552 - (-2002) = 1450 \text{ kJ mol}^{-1}$$

- When moving terms to the other side of the equation, the positive signs will become negative signs and vice versa
- **Remember:** Show your working, as instructed, so the examiner can see your thought process
 - If you make a mistake you can still gain marks for any correct further working

2c

c) There is no data for calcium oxide because:

- Calcium oxide reacts with water to produce OH^- ions in solution; [1 mark]

[Total: 1 mark]

- CaO (s) dissolves in water to form NaOH (aq)
 - $\text{CaO (s)} + \text{H}_2\text{O (l)} \rightarrow \text{Ca(OH)}_2 \text{ (aq)}$
 - This solution has a pH of around 12 (depending on the concentration)
- Therefore this does not fit in with the definition of enthalpy of solution:
 - The enthalpy change of solution is the enthalpy change when 1 mole of an ionic substance dissolves in water to give a solution of infinite dilution of the ions

3a

a) The equations for 1, 2 and 3 are:

- 1: $2\text{Na (s)} + \text{Se (s)}$; [1 mark]
- 2: $2\text{Na (g)} + \text{Se (g)}$; [1 mark]
- 3: $2\text{Na}^+ \text{ (g)} + \text{Se}^- \text{ (g)} + \text{e}^-$; [1 mark]

[Total: 3 marks]

- You must include electrons and state symbols when constructing Born-Haber cycles

3b

b) The change in lattice energy would be:

- 2: $2\text{Na (g)} + \text{Se (g)}$; [1 mark]
- 3: $2\text{Na}^+ \text{(g)} + \text{Se}^{2-} \text{(g)} + \text{e}^-$; [1 mark]

[Total: 3 marks]

- You must include electrons and state symbols when constructing Born-Haber cycles

3b

b) The change in lattice energy would be:

- Become more negative / become more exothermic; [1 mark]
- The S^{2-} ion is smaller than the Se^{2-} ion
OR
The S^{2-} ion has fewer electron shells than the Se^{2-} ion; [1 mark]
- Therefore stronger attraction between Na^+ ion and S^{2-} ion; [1 mark]

[Total: 3 marks]

- Lattice energy is the **formation** of 1 mole of an ionic lattice from gaseous ions under standard conditions
- The size of the ions is important when it comes to determining the difference in lattice energy
 - Formation is always **exothermic** as the lattice is made
- If the ions are smaller the attraction between the ions is stronger therefore the value for lattice energy is more negative / more exothermic

3c

c) To calculate the lattice energy:

- $\Delta H^\theta_f \text{Al}_2\text{O}_3 = 2\Delta H^\theta_{att}(\text{Al}) + 3\Delta H^\theta_{att}(\text{O}) + 2\Delta H^\theta_{ie1}(\text{Al}) + 2\Delta H^\theta_{ie2}(\text{Al}) + 2\Delta H^\theta_{ie3}(\text{Al}) + 3\Delta H^\theta_{EA1}(\text{O}) + 3\Delta H^\theta_{EA2}(\text{O}) + \Delta H^\theta_{latt}(\text{Al}_2\text{O}_3)$; [1 mark]
- $-1670 = (2 \times 326) + (3 \times 249) + (2 \times 578) + (2 \times 1817) + (2 \times 2745) + (3 \times -141) + (3 \times 753) + \Delta H^\theta_{latt}(\text{Al}_2\text{O}_3)$
 $-1670 = 652 + 747 + 1156 + 3634 + 5490 + (-423) + 2259 + \Delta H^\theta_{latt}(\text{Al}_2\text{O}_3)$
 $\Delta H^\theta_{latt}(\text{Al}_2\text{O}_3) = -1670 - 652 - 747 - 1156 - 3634 - 5490 + 423 - 2259$; [1 mark]
- $\Delta H^\theta_{latt}(\text{Al}_2\text{O}_3) = -15185 \text{ (kJ mol}^{-1}\text{)}$; [1 mark]

[Total: 3 marks]

- This calculation requires some thought given that the formula for aluminium oxide is Al_2O_3
- Therefore values in the table will have to be multiplied by either 2 or 3
- The enthalpy change for the formation of aluminium oxide does not need to be multiplied
 - $2\text{Al}(\text{s}) + 1.5\text{O}_2(\text{g}) \rightarrow \text{Al}_2\text{O}_3(\text{s})$
- Take your time with this and carefully go through each step making sure you have understood the change and then multiplied the value if required
 - For example, there are three oxygen atoms required, therefore the atomisation of oxygen electron affinity values must be multiplied by 3

4a

a) To identify M^+

- $\Delta H^\theta_{\text{sol}} = \Delta H^\theta_{\text{latt}} + (\Sigma \Delta H^\theta_{\text{hyd}})$; [1 mark]
- $+4 = 773 + (-363) + \Delta H^\theta_{\text{hyd}}\text{M}^+$
 $\Delta H^\theta_{\text{hyd}}\text{M}^+ = 4 - 773 - (-363)$
 $\Delta H^\theta_{\text{hyd}}\text{M}^+ = -769 - (-363)$
 $\Delta H^\theta_{\text{hyd}}\text{M}^+ = -406 \text{ kJ mol}^{-1}$ **AND** Na^+ ; [1 mark]

[Total: 2 marks]

- **Remember:** $\Delta H^\theta_{\text{hyd}}$ is for both ions that make up the ionic compound
- You then just need to rearrange the equation to find $\Delta H^\theta_{\text{hyd}}\text{M}^+$ and check this against the value in the table

4b

b) A possible formula for the new compound is:

- $\text{MBr}(\text{s}) / \text{MI}(\text{s})$; [1 mark]
- New ionic lattice has a less negative enthalpy of lattice formation
OR
New enthalpy of lattice formation is less exothermic; [1 mark]
- Therefore, the new halide ion must be bigger than the Cl^- ion
OR

Therefore the new halide ion must have more electron shells/ larger ionic radius than the C^- ion; [1 mark]

- So the attraction between the M^+ ion and the new halide ion is weaker than M^+ and the C^- ion; [1 mark]

[Total: 4 marks]

- The size of the ions matters!
- The larger the ion the greater the distance between the centre of the oppositely charged ions
 - This means the attraction between the ions will be weaker
- If the ions are bigger, the enthalpy of lattice formation is less exothermic or less negative

4c

c)

i) Enthalpy of hydration of Group 1 ions is negative:

- Water is polar / the oxygen in water has $\text{O}^{\delta-}$ charges; [1 mark]
- Group 1 ions / M^+ attract the $\text{O}^{\delta-}$; [1 mark]
- Bonds formed between ions and polar water molecules; [1 mark]

ii) The enthalpies of hydration become less negative down the group:

- The size of the ions increases down the group

OR

The number of electron shells increases down the group; [1 mark]

- The attraction between the positive Group 1 ion and the $\text{O}^{\delta-}$ charges decreases down the group; [1 mark]

[Total: 5 marks]

- Hydration enthalpy is a measure of the energy released when attractions are set up between positive or negative ions and water molecules
- The size of the hydration enthalpy is governed by the amount of attraction between the ions and the water molecules